BIOMETRIC SMARTCARD

UBIVELOX

UBIVELOX has strong technology and accumulated experience in the area of mobile widget and web service, solutions and service platforms for telematics and mobile. In addiction, UBIVELOX is the strong and competent company that has developed and supplied competitive smart card chips in various areas including finance, communication, transportation and public services as well as integrated smart card solutions to financial institution first in Korea, based on original technology of smart card areas.

FPC Manager TOOL User Guide V1.3

Introduction



The figure above shows the first time the **FPC Manager TOOL** is run.

This document introduces the **Fingerprint Enrollment & Verification** through the **FPC Manager TOOL**.

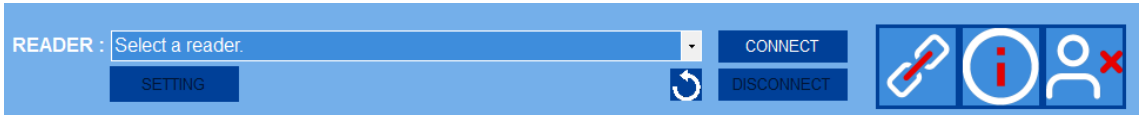
In order to register one user finger on a fingerprint card, a total of 16 fingerprint templates must be registered through the enrollment process and matching process.

1. In enrollment process :
   Register **8 fingerprint templates**
2. In matching process :
   Register an **additional 8 fingerprint templates**

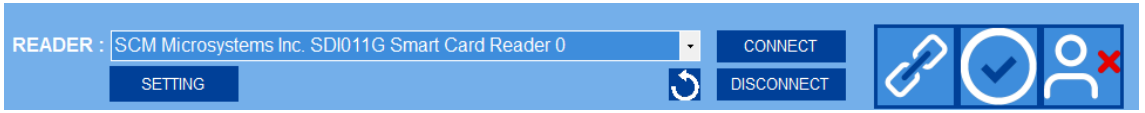
The user must register a total of 16 fingerprint templates and then proceed with the test.

Note) If you are using "FPC Manager TOOL" in a WINDOWS environment,
we is recommended not to use the "Certificate Propagation" service among WINDOWS service items.

Card Activation







Step.1

Choose a reader

Caution : Please place the card on the reader.
Do not remove the card from the reader
until fingerprint enrollment & verification is complete.

Step.2

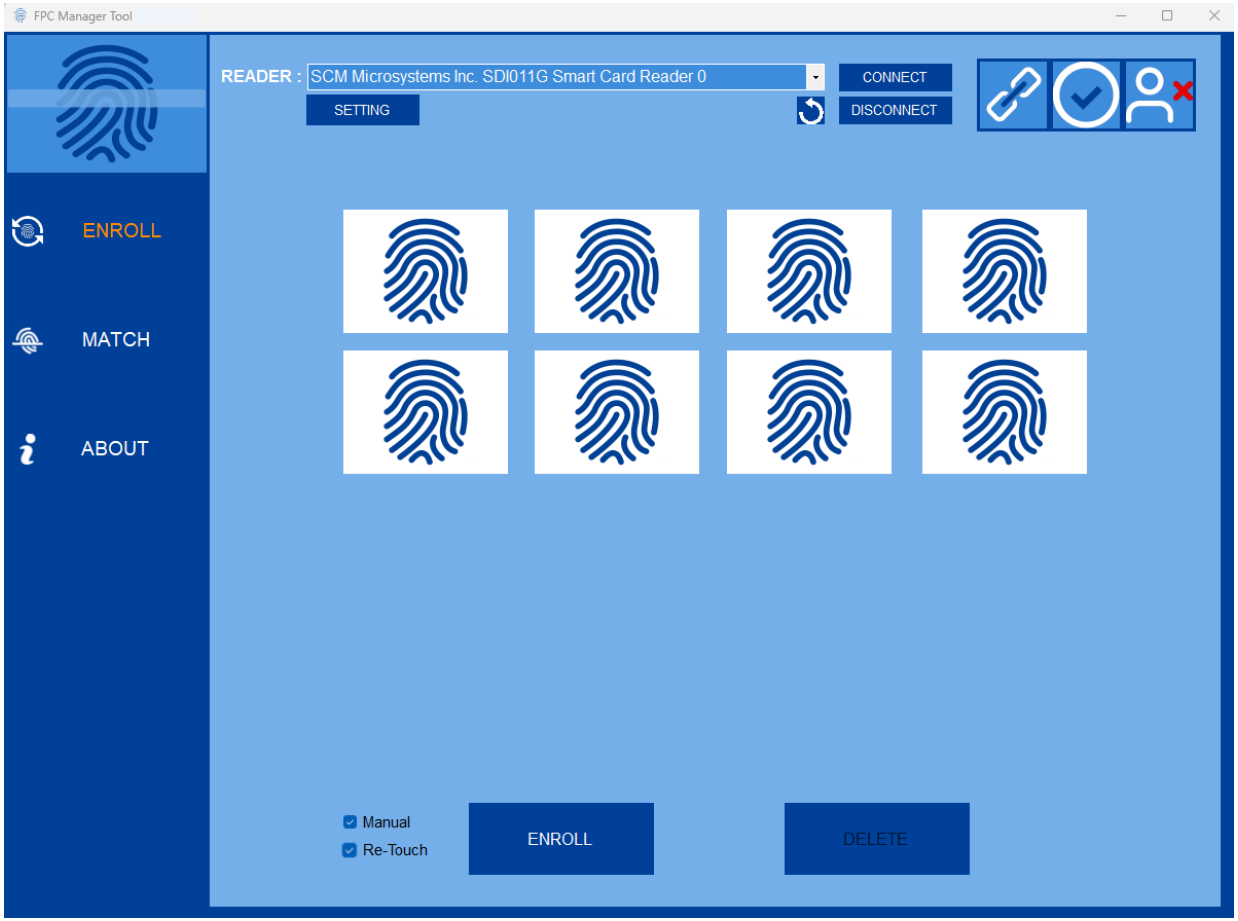
Click the "**CONNECT**" button

Step.3

When the reader is connected,
it will look like the picture on the right.



Enrollment (1/7)



In enrollment process :

Register **8 fingerprint templates**

Two mode (to register the fingerprint template)

- Manual Mode : ☑ Manual (Checked)
- Auto Mode : ☐ Manual (Un-Checked)

Manual mode

You can register only one fingerprint template by clicking the Enroll button in manual mode.

Auto mode

You can register multiple fingerprint templates (8 templates) by clicking the Enroll button in auto mode.

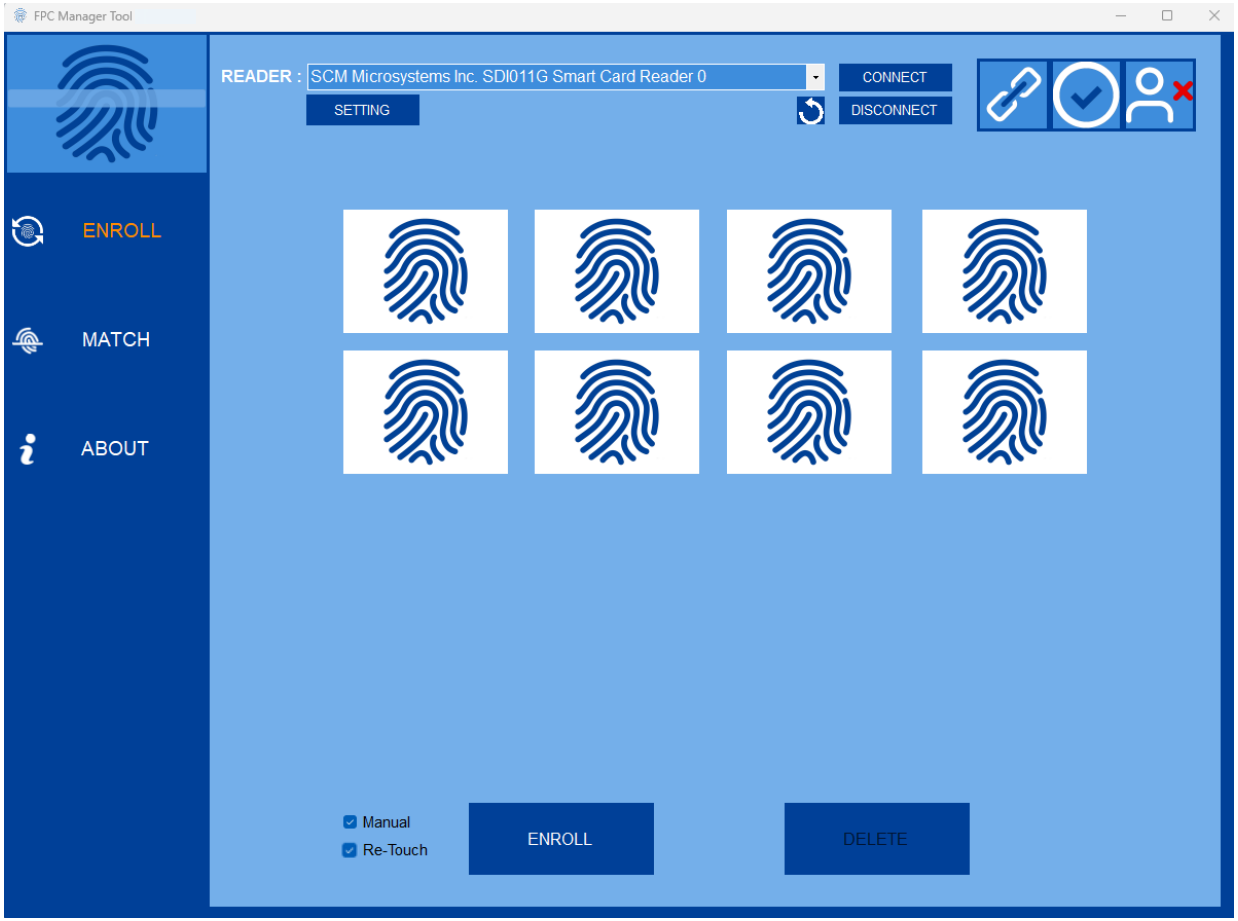
Re-Touch Function

When the Re-Touch checkbox is selected, you must remove your finger from the card sensor and place your finger on the card sensor again after each fingerprint template registration.

Caution) **The checkbox must be checked only if the card provides the Re-Touch function.**

Please contact Ubivelox to check whether the card provides the Re-Touch function.

Enrollment (2/7)



In enrollment process :

Register **8 fingerprint templates**

The figure below shows the process of registering fingerprint template in **three steps**.



Status A) Initialization status



Status B) Template extractable state



Status C) Template extraction completion status

Enrollment (3/7)



In enrollment process :

Register **8 fingerprint templates**

Step.1
Click the “ENROLL” button to register your fingerprint.

Step.2
When the fingerprint Enroll status image changes from “Status A” to “Status B”, **place the finger (to be registered) on the sensor.**

Step.3
When the fingerprint Enroll status image changes from “Status B” to “Status C”, **remove the finger on the sensor.**

[In Manual mode]
Repeat steps 1-3. (**Until** 8 fingerprint images are changed to Status C)

[In Auto mode]
Repeat steps 2-3. (**Until** 8 fingerprint images are changed to Status C)



Enrollment (4/7)



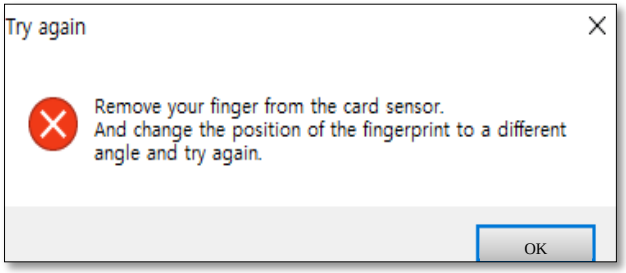
In enrollment process :

Register **8 fingerprint templates**

Caution 1:

If fingerprint registration is not performed well,

please refer to the **Enrollment Guide** on page 11.



If fingerprint registration is not performed well,

a pop-up window like the one above may be displayed.

When this pop-up window appears,

click the OK button and proceed **again**.

Enrollment (5/7)



In enrollment process :

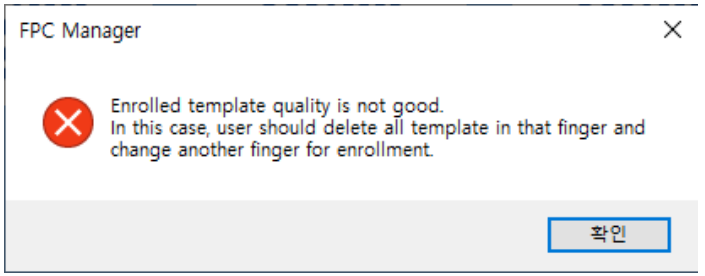
Register **8 fingerprint templates**

Caution 2:

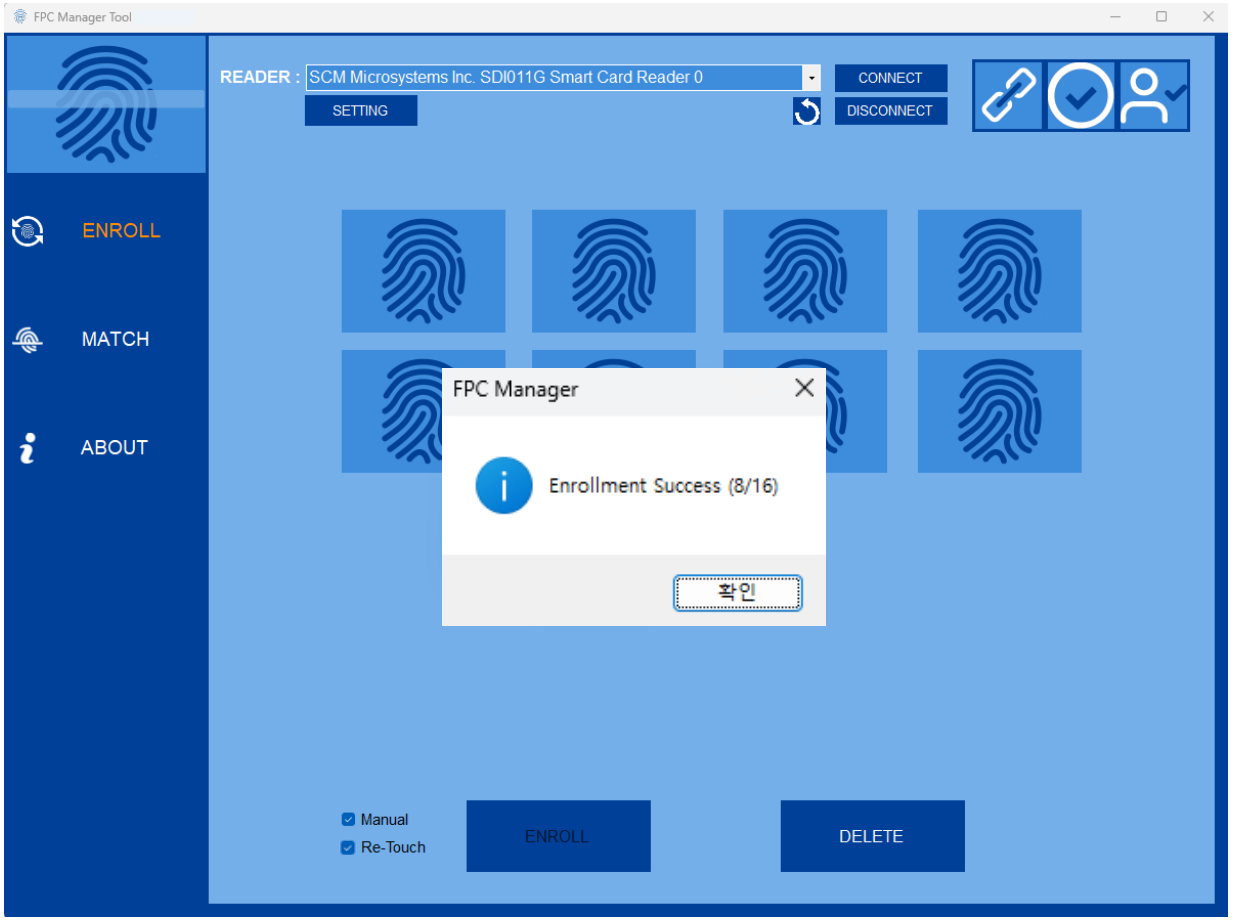
When the Enrolled template quality is not good, the below pop-up window appears.

In this case,

user should delete all template in that finger and change another finger for enrollment.



Enrollment (6/7)



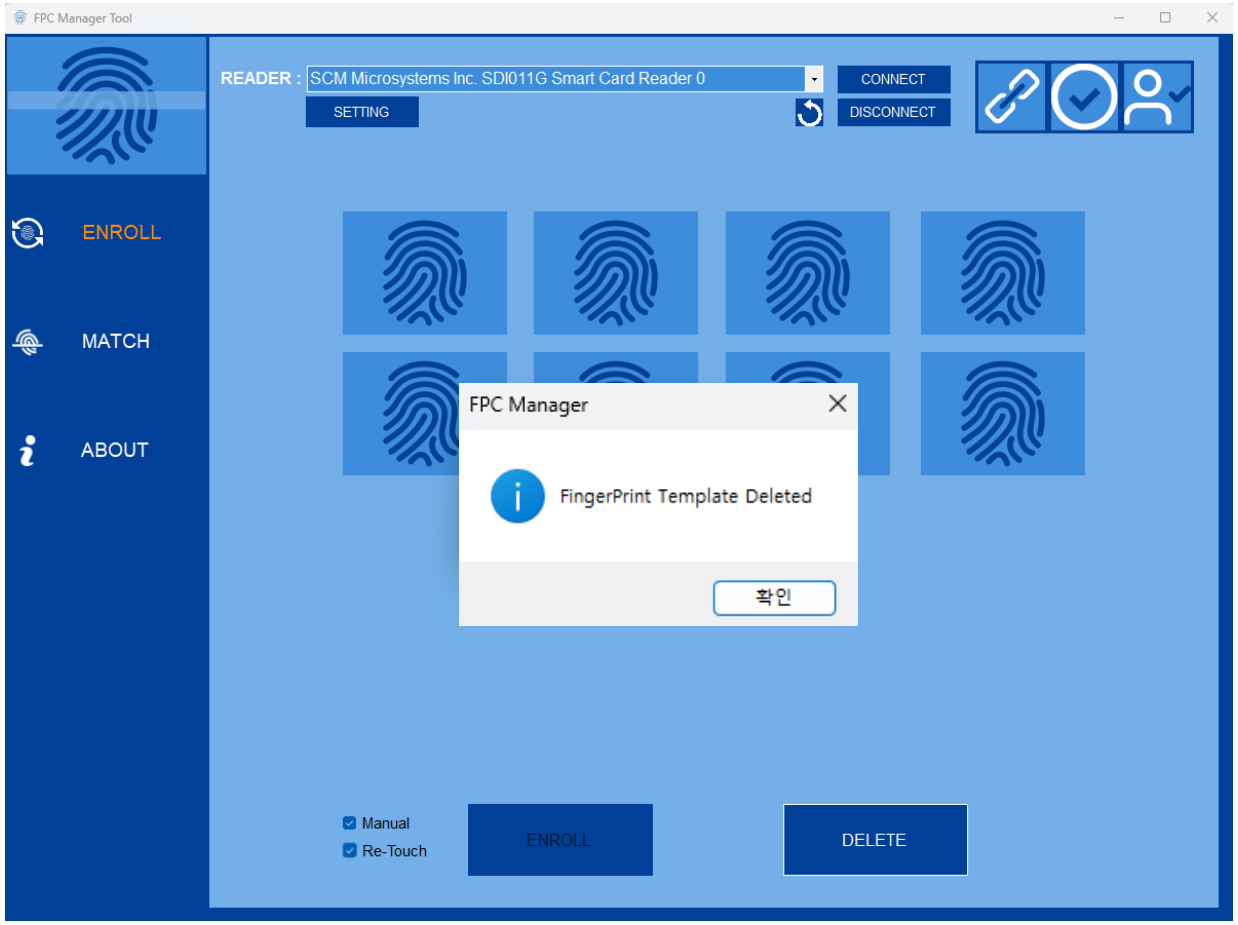
If fingerprint template registration is complete,

An " **Enrollment Success**" pop-up window is displayed.

In enrollment process :

Register **8 fingerprint templates**

Enrollment (7/7)



In enrollment process :

Register **8 fingerprint templates**

How to delete registered fingerprint?

If you press the “**DELETE**” button,
the previously registered fingerprint is deleted.

When the fingerprint template deletion is complete,
a “**Fingerprint Template Deleted**" pop-up window will appear.

Enrollment Guide

Please refer to table below.

Template Number	1	2	3	4
Angle	about -45°	about -30°	about -15°	about -10°
Example				

Template Number	5	6	7	8
Angle	about -5°	about 0°	about 5°	about 10°
Example				

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The fingerprint template should be stored a total of **8 templates per finger**. Each of the eight templates should be stored at a different angle and various sides of fingerprint.

If more than **4 poor quality templates** are enrolled, user cannot no longer register fingerprint. In this case, user should delete all templates of the finger and proceed with registration again.

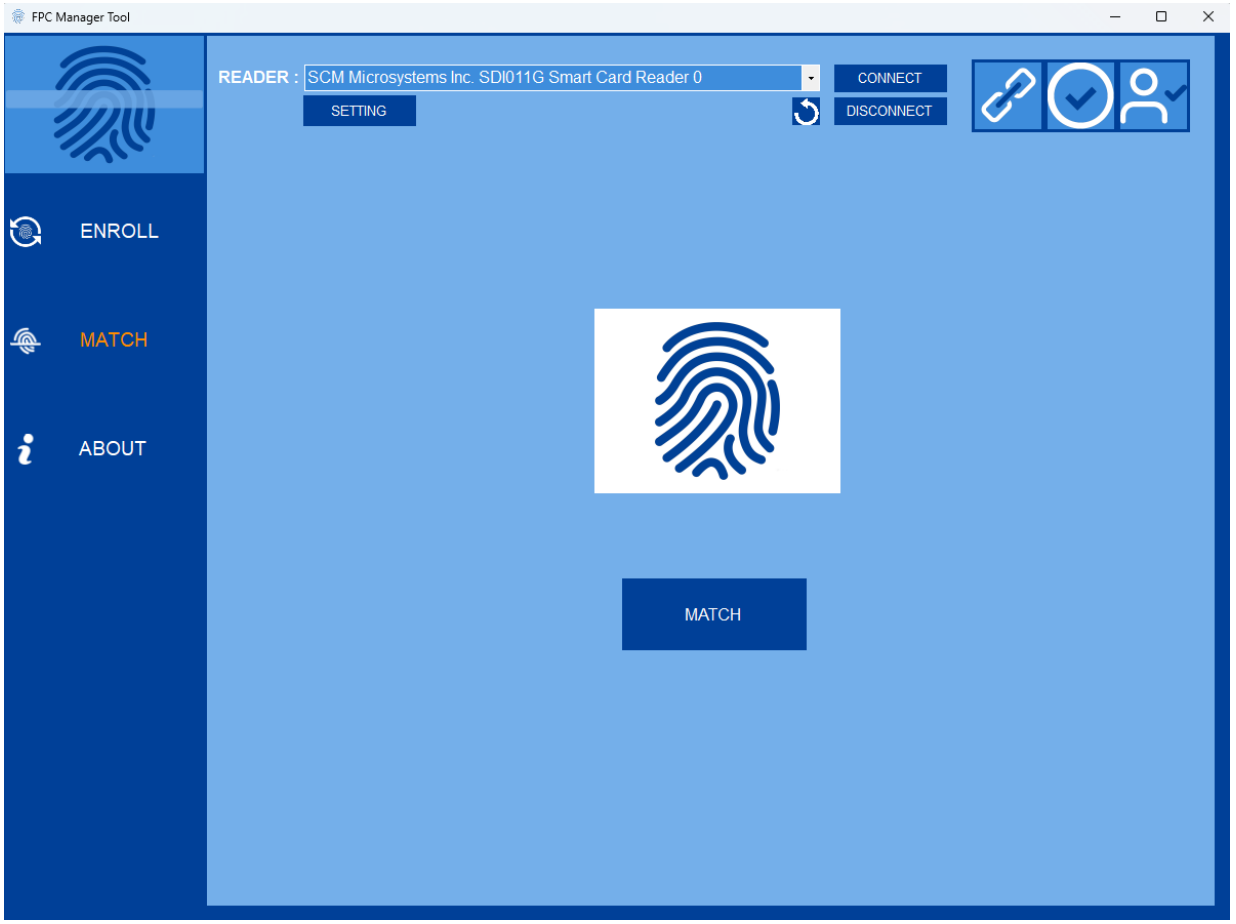
For example**, poor quality template** may be stored in the following cases.

- Fingerprints wet with sweat or water.
- Fingerprints with a low difference in the protuberance
- Fingerprints captured only in part clearly.

NOTE :

In order to guarantee performance of fingerprint library, more than **70%** of fingerprint

Matching



In matching process :

Register **additional 8 fingerprint templates**

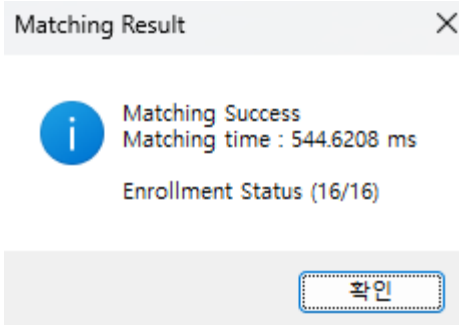
(Repeat fingerprint matching until 16 fingerprint templates are registered.)

Step.1

Place your finger on the sensor and press the "**MATCH**" button.

Step.2

If fingerprint template matching is complete, An " **Matching Result**" pop-up window is displayed.

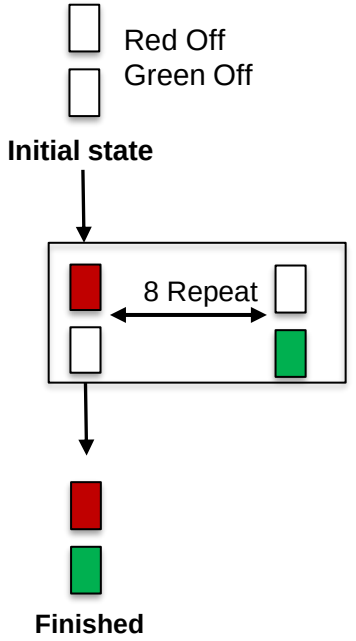


[Matching Success pop-up window]

Note) The matching time measured in PC TOOL may be longer than measured time in CARD.

LED Status

Please reference this page only if the card provides LED.

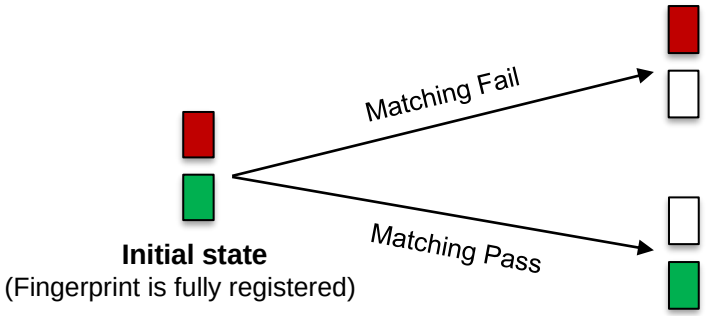


Enrollment

Case	Green	Red	Explain
1	OFF	OFF	Initial state (Without fingerprint registration)
2	OFF	ON	Template extractable state (User Action : Put your finger on the sensor)
3	ON	OFF	Template extraction completion status (User Action : Take your finger off the sensor)
4	ON	ON	Fingerprint is fully registered (for 8 templates)

Matching

Case	Red	Red	Explain
1	OFF	OFF	Matching Initial state (Without fingerprint registration)
2	ON	ON	Matching Initial state (Fingerprint is fully registered)
3	OFF	ON	Matching : **Fail** (or Not Attempt)
4	ON	OFF	Matching : **Pass**



Revision history

Date	version	Changes	Author
2022.03.02	1.0	Initial release	Ubivelox
2022.05.24	1.1	Add Manual Mode	Ubivelox
2022.08.30	1.2	Change the enrollment count of fingerprint template	Ubivelox
2025.03.18	1.3	Add Re-Touch Function	Ubivelox


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